

THE ALBERTA DISABILITY SYSTEM BREAKDOWN · DOCUMENTATION SERIES

WHAT'S ON THE SHELF

Income, Exposure, and the Disability Community's Food Supply

MAY 2026 EDITION

A documentation piece on the structural concentration of food-supply exposure burden in the disability cohort the AISH and ADAP framework defines

The Alberta Disability System Breakdown

St. Albert, Alberta · May 2026

albertadisabilitybreakdown@outlook.com

albertadisabilitysystembreakdown.netlify.app

About this document

What's On The Shelf is a documentation piece produced by The Alberta Disability System Breakdown, an independent public advocacy campaign documenting the mandatory transition of approximately 79,290 Albertans from the Assured Income for the Severely Handicapped (AISH) program to the Alberta Disability Assistance Program (ADAP), effective July 1, 2026.

The campaign operates as a documentation-first public record. It is not a support group, a service provider, or a political advocacy organization in the conventional sense. Its work is to assemble, source, and make publicly available the legislative, fiscal, regulatory, and structural record of the AISH-to-ADAP transition and the policy framework within which it has been undertaken.

This piece extends the campaign's structural analysis to the food supply. The argument it develops is narrow: that the income levels at which AISH and ADAP households are required to live force consumption patterns that concentrate exposure to documented contaminants — heavy metals in rice and rice-based products, synthetic dyes regulated more strictly in comparable jurisdictions, additives associated in the peer-reviewed literature with neurodevelopmental and behavioural effects — in the disability cohort the income framework defines. The argument is not that processed food causes disability. The argument is that disability income policy structurally allocates exposure burden to the population it serves.

Citations are provided throughout. The principal academic, regulatory, and primary-source references are listed in the consolidated bibliography. The piece may be freely quoted, reproduced, and redistributed under the terms of attribution to The Alberta Disability System Breakdown. No individual byline is asserted. The campaign holds the work as collective record.

Suggested citation: The Alberta Disability System Breakdown, *What's On The Shelf: Income, Exposure, and the Disability Community's Food Supply*, St. Albert, May 2026, albertadisabilitysystembreakdown.netlify.app.

The shelf is what is on the shelf. The question of who lives on what the shelf contains is a question of income, not of choice.

A Note on This Work

This piece is part of a longer documentation series produced by The Alberta Disability System Breakdown across 2025 and 2026. The income framework it draws on is documented at length in the campaign's *Income Crossover Analysis* (revised May 2026), *The Cage Doesn't End* compilation (May 2026), and *Building Up: A Blueprint for the Alberta Accessibility and Inclusion Act* (May 2026). The regulatory architecture under which the food supply is approved is documented in the Food and Drug Regulations, in Health Canada's permitted-additives and maximum-levels lists, in the Canadian Food Inspection Agency's surveillance publications, and in the comparator regulatory records of the European Union and the United States Food and Drug Administration.

What this piece is not. It is not a piece about food choices. The disabled Albertan whose monthly income after rent leaves between forty and four hundred forty dollars for food, utilities, transportation, medical out-of-pocket expenses, and household supplies is not choosing the rice-based diet, the shelf-stable processed-food diet, the food-bank-supplemented diet. The economic structure is choosing it. The piece develops the documented record of that structural choice.

It is not a piece that links processed food to autism, to ADHD, or to any specific developmental outcome. That literature exists; the campaign is not the appropriate vehicle for it. The exposures documented here — arsenic, cadmium, lead, synthetic dyes, sodium benzoate, and the broader list — are documented as exposures, against documented regulatory standards in comparator jurisdictions. The structural argument concerns the income-driven concentration of exposure in the disability cohort, not the population-level causation of any specific diagnosis.

It is not a piece that names individual policy designers, food regulators, or manufacturers as having acted in bad faith. The constructive knowledge doctrine the piece engages attaches to the institutional record — to what governments and regulatory agencies have on their own publications, what they have on their own consultation records, and what they have on their own surveillance data — not to the personal awareness of any individual actor.

Audience. The intended audience is institutional — oversight bodies, public health researchers, journalists, elected representatives willing to engage — together with disability community members who have lived the structural condition this piece names and who have asked, often, whether the conditions of their household food supply are conditions they have produced or conditions that have been produced for them. The piece argues, from the documented record, that the answer is the second.

Contents

FRONT MATTER

About this document	ii
Epigraph	iii
A Note on This Work	iv
Contents	v

PART ONE — THE INCOME CONSTRAINT

I. The Income Constraint: What AISH and ADAP Households Can Afford	1
--	---

PART TWO — WHAT IS ON THE SHELF

II. Rice on the Shelf: Heavy Metals and the Cheapest Staple	7
III. The Colours of Convenience: Synthetic Dyes Permitted in Canada	13

PART THREE — THE STRUCTURAL CONCENTRATION

IV. The Food Bank as Symptom: Disability and Hunger in Alberta	19
V. The Compound Effect: Exposure Burden Concentrated in the Disability Cohort	25

CLOSING

What This Establishes	31
Bibliography and Sources	33

This is the May 2026 edition. The full source apparatus is available in the bibliography at the close and at albertadisabilitysystembreakdown.netlify.app.

PART ONE · THE INCOME CONSTRAINT

Chapter I

The Income Constraint

What AISH and ADAP Households Can Afford

The argument this piece develops begins with a number. The maximum monthly living allowance under the Assured Income for the Severely Handicapped program in 2026 is \$1,940. The maximum monthly living allowance under the Alberta Disability Assistance Program, which receives the transitioning AISH caseload on July 1, 2026, is \$1,740. Recipients of the transition payment retain the AISH rate until December 31, 2027; on January 1, 2028, the payment drops to the ADAP rate. These are the income figures the policy framework has set.¹

The number matters because food does not arrive on the disabled household's table by accident. It arrives through the constrained budget of a recipient whose income the same provincial government that set the food-safety framework also set. The income figure determines what the household can buy. The food regulator determines what is on the shelf. The disabled recipient lives inside the intersection. This chapter documents the intersection.

Rent First

The Canada Mortgage and Housing Corporation reports average one-bedroom market rent in Edmonton at approximately \$1,500 per month and in Calgary at approximately \$1,900 per month in 2024, with documented year-over-year increases of 17.5 percent province-wide. The average rent across all unit sizes in Alberta was \$1,792 in 2024; three-bedroom averages reached \$2,138, with documented increases of 21.3 percent year-over-year in that segment. Alberta does not impose a residential rent-increase cap.²

The arithmetic produces the documented condition the campaign's prior analyses have named the structural cage. A single AISH recipient on the maximum living allowance of \$1,940 paying \$1,500 in rent retains \$440 per month for everything else. A single AISH recipient paying \$1,900 in rent — the upper end of the one-bedroom range — retains \$40. A single ADAP recipient, post-transition, paying the lower-end rent of \$1,500 retains \$240. A single ADAP recipient paying the upper-end rent retains a negative balance: the rent exceeds the benefit by \$160 per month before any other expense.³

The residual figure of \$40 to \$440 is the figure that the rest of the household's monthly budget — food, utilities, transportation, medical out-of-pocket expenses, clothing, household supplies, personal-care items, household goods — must come out of. The figure is the operational constraint that determines what enters the household's grocery cart, what does not, what is purchased at the grocery store at all, and what is obtained from the food bank, the church basement, the community kitchen, or the household budget of a family member who is themselves not on a disability benefit.

The Cost of Food

Statistics Canada estimates the average monthly food cost for a single adult aged 31 to 50 in 2026 at between \$370 and \$460 depending on the food-cost basket applied. The Canada Food Price Report 2026, produced by Dalhousie University in collaboration with the University of British Columbia, the University of Saskatchewan, and the University of Guelph, projects an additional food-cost increase of three to five percent over 2025 levels, building on the 23 percent cumulative food-price increase across Canada since February 2020.⁴

The figures presented in the previous paragraph are the figures the average Canadian adult faces. They are not adjusted for the additional dietary requirements common in the AISH and ADAP populations: prescribed dietary modifications for diabetes (commonly co-occurring with mental-health and cognitive presentations and documented in approximately twenty percent of the AISH caseload), gluten or dairy intolerance (substantially elevated in the autism-spectrum cohort), the higher protein and calorie requirements associated with several chronic-pain and fatigue-related conditions, the prepared and texture-modified food requirements associated with severe dental and oral-motor presentations the population also exhibits. The Statistics Canada figure is the floor, not the median household experience.

Against the figure of \$40 to \$440 retained after rent, the food cost is the largest single non-rent expense the household must absorb. It is not a flexible budget item. The household that goes without food is the household whose recipient does not, on the documented evidence base in the peer-reviewed nutritional literature, sustain the medical condition the disability benefit was designed to support. The food budget is the operational floor of the disability framework. The framework's income structure does not meet the floor.⁵

Utilities, Transportation, Medical

Alberta utility estimates for a single-occupant residential dwelling in 2025-2026 range from approximately \$385 to \$510 per month. The range reflects substantial regional variation: the figures are higher in rural Alberta where natural-gas heating is dominant and lower in the central urban corridor where district-heating and smaller-footprint dwellings reduce the per-recipient cost. The figure does not include the additional electrical load common in households where a recipient uses powered medical equipment, oxygen concentrators, sleep-apnea machines, or powered-mobility devices that charge overnight.⁶

Transportation costs are not optional in a province in which medical appointments, specialist visits, pharmacy pickups, and the network of supports the disability framework requires the recipient to navigate are distributed across multiple physical locations. Edmonton Transit Service and Calgary Transit both offer reduced monthly pass programs for AISH recipients; the reduced rates remain substantial against the residual income figure. Recipients outside the two metro centres pay the full provincial average. Specialized transportation (DATS in Edmonton; Access Calgary) requires advance booking and operates with documented capacity constraints.⁷

Medical out-of-pocket expenses are the category of expense the income framework least anticipates and the disability population most consistently faces. The Disability Tax Credit physician certification fee, the prescription dispensing fee in the post-pharmicare landscape, the dental and vision expenses not covered by the Alberta Adult Health Benefit, the medical-device replacement and battery costs, the non-prescription supplement and over-the-counter expenses many specialist physicians recommend without formal prescription

— the cumulative monthly expense in this category is documented in the campaign's *Access Barrier Report* at between \$100 and \$400 for a typical AISH or ADAP household.⁸

What Is Left for the Grocery Cart

The arithmetic produces a documented residual. After rent at the lower end of the one-bedroom range, utilities at the lower end of the Alberta range, transportation at the reduced AISH transit-pass rate, and medical out-of-pocket expenses at the lower end of the typical-household range, the AISH recipient retains approximately \$95 to \$300 per month for food and all other household expenses. The ADAP recipient, post-January 2028, retains approximately negative \$105 to \$100 per month for the same categories. The residual is, in many configurations, negative before food enters the calculation.⁹

The household that retains \$300 per month for food can, on the Statistics Canada figure, purchase approximately seventy-five percent of the food a typical single adult is expected to consume in a month. The household that retains \$95 per month for food can purchase approximately twenty-five percent. The household that retains a negative balance cannot purchase food at all from the residual; the food is obtained, where it is obtained, through the food bank, through familial or community subsidy, or through the reduction of one of the other expense categories — typically utilities (the recipient cuts heat in winter), medical (the recipient defers a prescription refill or a specialist appointment), or transportation (the recipient misses a medical appointment).¹⁰

The choices the structural framework forces are not random. They are the choices that produce the food-bank-reliance pattern documented in Chapter IV, the medication-non-adherence pattern documented in the broader Alberta chronic-disease literature, and the consumption pattern — heavy reliance on rice, on shelf-stable processed foods, on the cheapest available staples — that determines what enters the chapters that follow.

The Shelf the Household Reaches

The grocery shelf the disabled household reaches with the residual figure documented above is not the full grocery shelf. It is the bottom-cost segment of the grocery shelf. The bottom-cost segment is, in the documented Canadian food market, the segment in which the cheapest sources of carbohydrate are rice and rice-based products (bagged white and brown rice; rice-based pasta and noodle products; rice-based shelf-stable meals; rice-flour-based bread and bakery products in the gluten-free aisle the recipient with celiac disease must purchase from). It is the segment in which the cheapest sources of protein are processed-meat products in shelf-stable packaging, often containing the synthetic dyes documented in Chapter III. It is the segment in which the cheapest sources of fruit and vegetable intake are canned (with the higher sodium and additive load of the canning process), frozen (with the food-dye load of the frozen-prepared-meal product line), or pre-packaged shelf-stable (with the full documented additive and dye load of the processed-food category).¹¹

The household that retains \$40 to \$440 per month for food and household expenses is not making the choice between organic produce and conventional produce, between the higher-quality brand and the lower-quality brand, between the fresh-prepared meal and the shelf-stable meal. The household is making the choice between the cheapest available items and going without. The cheapest available items are, on the documented evidence base the chapters that follow assemble, the items with the highest exposure burden in the

contemporary Canadian food supply.

What This Chapter Establishes

Three observations follow from this chapter, and the chapters that follow draw on them without restating.

First, the AISH and ADAP income figures, against the documented Alberta rent, utility, transportation, and medical cost structure, produce a residual food and household budget that is between zero and approximately thirty percent of the food cost a typical single adult is expected to absorb. The residual is not a budgeting challenge for the recipient. It is a structural feature of the income framework.

Second, the residual figure determines the grocery shelf the disabled household reaches. The shelf the household reaches is the bottom-cost segment, in which the documented exposure burden is concentrated. The household is not making the choice to purchase the higher-exposure items. The household is making the choice between the cheapest available items and going without food.

Third, the income framework that produces the constraint is the framework operated by the same provincial government that operates the food-safety regulatory framework the chapters that follow document. The two frameworks are not separately optimized for unrelated purposes. They operate, on the documented record, on the same population. The chapters that follow document what the food-safety framework permits on the shelf the income framework has restricted the disabled household to.

Notes — Chapter I

1. Government of Alberta, AISH and ADAP program documentation, 2025-2026; Assured Income for the Severely Handicapped General Regulation, AR 96/2026, filed May 12, 2026, in force July 2, 2026. The Alberta Disability System Breakdown, *Income Crossover Analysis*, revised May 2026, for full worked income trajectories.
2. Canada Mortgage and Housing Corporation, Rental Market Report 2024; Urbanation/Rentals.ca, National Rent Report, June 2024 (17.5 percent Alberta year-over-year increase); GIL Property Management, "Alberta Rent Prices in 2024," November 2024 (three-bedroom average \$2,138/month, 21.3 percent year-over-year increase). Alberta does not impose a residential rent-increase cap.
3. Arithmetic calculations from the published AISH (\$1,940) and ADAP (\$1,740) base rates against the CMHC and Urbanation rent figures. See *Income Crossover Analysis* (revised May 2026) for full worked calculations across employment-income, household-composition, and transition-payment scenarios.
4. Statistics Canada, monthly food cost estimates for a single adult age 31-50, 2026; Dalhousie University, University of British Columbia, University of Saskatchewan, and University of Guelph, *Canada's Food Price Report 2026*, December 2025. Cumulative food-price increase of 23 percent since February 2020 reported by Statistics Canada Consumer Price Index data.
5. Peer-reviewed nutritional literature on disability and chronic illness dietary requirements is extensive; see, e.g., the Academy of Nutrition and Dietetics' position papers on intellectual disability, on autism spectrum disorder, and on chronic-pain dietary management for primary-source synthesis. The AISH caseload primary-medical-condition breakdown is from Government of Alberta, AISH Caseload Open Data, September 2025.
6. Alberta utility cost estimates compiled in *The Alberta Disability System Breakdown*, AISH Evidentiary Report Series (April 2026), drawing on Government of Alberta consumer-utility data, regional cost-of-living surveys, and recipient-level reports. The estimates do not include the elevated electrical load associated with powered medical equipment.
7. Edmonton Transit Service and Calgary Transit reduced-pass program documentation, 2025-2026; DATS and Access Calgary specialized transportation documentation; Government of Alberta Provincial Bus Pass program documentation. The reduced-pass rates are documented but remain a meaningful expense against the income figures established in this chapter.

8. The Alberta Disability System Breakdown, *Access Barrier Report*, April 2026, with primary-source receipts and timestamped correspondence spanning 2019-2026. See also *The Access Gap: Penalized for Non-Ability*, May 2026, on the structural barriers to medical access in the AISH/ADAP cohort.
9. Worked residual calculations following the framework established in *Income Crossover Analysis* (revised May 2026). The negative-residual case is the documented condition of a substantial portion of the ADAP cohort under post-January 2028 conditions when the transition payment expires.
10. Calculations against Statistics Canada single-adult monthly food cost estimate of \$370-\$460 (2026). The food-bank-reliance pattern, the medication-non-adherence pattern, and the missed-medical-appointment pattern are documented in Chapter IV of this piece and in the broader Alberta chronic-disease and food-insecurity literature.
11. The bottom-cost grocery shelf is documented in the household-budget research literature; see, e.g., the Maytree Foundation's welfare-incomes research, the Cardus Family Office's grocery-basket affordability studies, and the academic literature on food insecurity in low-income households. The specific composition of the bottom-cost shelf in the Canadian market is the subject of Chapters II and III of this piece.

PART TWO · WHAT IS ON THE SHELF**Chapter II****Rice on the Shelf***Heavy Metals and the Cheapest Staple*

Chapter I established the income constraint. The household with \$40 to \$440 per month for food after rent does not have full access to the grocery shelf. It has access to the bottom-cost segment. The bottom-cost segment, in the contemporary Canadian food market, is anchored by rice and rice-based products. This chapter documents what is in those products.

The chapter's argument is narrow and structural. The contemporary Canadian food regulatory framework does not establish a maximum level for inorganic arsenic in infant rice cereal or in rice sold to the general consumer. The European Union established a 100 parts-per-billion limit for inorganic arsenic in rice products intended for infants in 2016. The United States Food and Drug Administration set a 100 parts-per-billion final guidance limit for inorganic arsenic in infant rice cereal in 2021. Health Canada announced a consultation toward a similar limit in 2019. As of the date of this edition, May 2026, that consultation has not produced a maximum level. The shelf in Canada is, on the regulatory record, less protected than the shelf in comparator jurisdictions.¹

The HBBF May 2025 Investigation

Healthy Babies Bright Futures is a coalition of scientists, non-profits, and donors that has, since 2017, published primary-source testing of heavy-metal contamination in the foods commonly fed to infants and young children. The coalition's 2017 study of 105 infant cereals found that infant rice cereal contained on average six times more inorganic arsenic than other infant cereals. The coalition's 2019 follow-up found that 4 of 7 infant rice cereal samples exceeded the 100 parts-per-billion limit subsequently adopted by the FDA. The coalition's broader 2019 testing of baby foods across the market found heavy metals — arsenic, cadmium, lead, or mercury — in 95 percent of 168 baby foods tested.²

On May 15, 2025, HBBF published its most extensive investigation to date. The study tested 145 rice samples purchased at retail stores in 20 metropolitan markets across the United States. The samples included U.S.-grown and imported rice across the major commercial brands available on North American retail shelves, including brands that distribute into the Canadian market. The headline finding: 100 percent of 145 rice samples contained arsenic. More than one in four samples exceeded the FDA's 100 parts-per-billion limit for infant rice cereal — a limit that, the study notes, does not apply to rice itself, only to rice cereal marketed for infants. Cadmium was found in all but one sample. Lead was elevated in seasoned rice products.³

The HBBF findings are not contested in the peer-reviewed scientific literature. The presence of inorganic arsenic in rice is a documented consequence of rice's agronomy: rice is grown in flooded paddies, the flooding

mobilizes arsenic from the surrounding soil and groundwater, and the rice grain accumulates the mobilized arsenic at concentrations approximately ten times higher than non-rice grains grown in the same regions. The accumulation is the documented mechanism. The dispute is not whether arsenic is present. The dispute is the regulatory question of what concentration is permitted on the consumer shelf and the public-health question of what cumulative exposure produces the documented downstream effects.⁴

The Documented Effects

The U.S. National Institute of Environmental Health Sciences (NIEHS), the U.S. Agency for Toxic Substances and Disease Registry, the World Health Organization, and Health Canada itself all classify inorganic arsenic as a Group 1 human carcinogen. The carcinogenic classification is not contested across the major regulatory and scientific bodies. The non-carcinogenic effects documented at lower chronic-exposure levels include neurodevelopmental impairment in exposed children, including documented IQ reduction in birth cohorts followed longitudinally; cardiovascular disease in adults; type 2 diabetes; and immune-system modulation.⁵

The New Hampshire Birth Cohort Study, the longest-running North American longitudinal cohort on infant arsenic exposure, has documented elevated urinary arsenic in infants whose first foods include rice cereal and rice-based products, with infants who consumed rice-based infant foods exhibiting urinary arsenic concentrations approximately three times higher than infants who did not. The Dartmouth Children's Environmental Health and Disease Prevention Research Center, which operates the cohort, has published the findings across multiple peer-reviewed outlets since 2016.⁶

The body-weight-adjusted exposure is the figure that matters most to the structural argument this piece develops. An infant or young child consuming rice or rice-based products consumes those products at a body-weight-adjusted rate approximately three times higher than an adult consuming the same products. The developmental window during which the central nervous system is most sensitive to neurotoxic exposure is the window from conception through approximately age six. The infant of a low-income household — including a household on AISH or ADAP — whose first solid food is infant rice cereal, whose second solid food is rice-based puffs and teething biscuits, whose family diet relies on rice as a staple because rice is the cheapest available carbohydrate on the bottom-cost shelf, is the infant whose body-weight-adjusted exposure is at the peak the cohort literature documents.⁷

The Canadian Regulatory Position

Health Canada's regulatory position on arsenic in food is documented in Part B of the Food and Drug Regulations and in the agency's Maximum Levels for Various Chemical Contaminants in Foods list. The list establishes maximum levels for arsenic in specific food categories: fish protein concentrate at 350 parts per billion; bottled water at 10 parts per billion (mirroring the World Health Organization drinking-water guideline). The list does not establish a maximum level for inorganic arsenic in rice, in rice-based infant cereal, or in rice-based consumer products.⁸

On April 4, 2019, Health Canada announced through media adviser Maryse Durette that the agency would launch a months-long consultation on setting a maximum level for arsenic in food, including baby cereal. The announcement followed a CBC Marketplace investigation that had identified levels of arsenic in baby foods sold in Canada which, under the 2016 European Union regulation, would have required the products to be removed from European shelves. The Health Canada announcement said the consultation would produce a

proposal "by mid-2019" and that the agency would "recommend new maximum levels for inorganic arsenic in rice, consistent with those established by the Codex Alimentarius Commission."⁹

The Codex Alimentarius Commission, the joint Food and Agriculture Organization and World Health Organization international standards body, adopted a maximum level of 100 parts per billion for inorganic arsenic in polished rice in 2014, and a maximum level of 200 parts per billion in husked rice in 2016. The Codex limit is the international reference standard. Health Canada's 2019 announcement committed to adoption consistent with the Codex standard. As of the date of this edition, May 2026 — seven years after the announcement — Health Canada has not adopted a maximum level for inorganic arsenic in rice, in infant rice cereal, or in any rice-based product.¹⁰

Alberta Health Services and the Constructive Knowledge Record

The constructive knowledge condition the campaign's documentation has engaged across its prior reports operates with particular clarity in the arsenic-in-rice case. Alberta Health Services maintains, as part of its Nutrition Guidelines for Healthy Infants and Young Children, a guideline document titled *Arsenic in Foods*. The document, prepared by AHS Nutrition Services and addressed to primary-care physicians, public-health nurses, and community-health workers, advises that "it is possible that infants and toddlers could be exposed to higher levels of inorganic arsenic from regular and frequent consumption of infant rice cereal and processed rice snack products," advises practitioners to counsel families to limit rice-based snack foods, and notes that "if Health Canada's proposal (to set more protective limits on the allowable amounts of inorganic arsenic in rice products intended for infants and young children) is implemented, rice-based complementary foods may no longer be viewed as potentially problematic."¹¹

The Alberta Health Services document is the constructive knowledge record. The same provincial government that has not, as of the date of this edition, ensured that Health Canada's promised consultation has produced a regulatory limit, has instructed its own public-health practitioners in writing to advise families that rice-based foods pose a documented arsenic exposure concern. The instruction is the documented institutional position. The regulatory limit that would address the exposure on the shelf is, on the documented record, not in place.

The structural effect is the effect this chapter and the chapters that follow document. The disabled household, restricted by the income framework to the bottom-cost grocery shelf, purchases rice and rice-based products at a rate higher than households with the income flexibility to choose alternative carbohydrate sources. The infant in that household, the toddler in that household, the AISH or ADAP recipient in that household whose primary diagnosis includes the cognitive or mental-health presentations documented in the AISH caseload data — that household, in its disproportionate consumption of the rice-based food category, is the household at which the documented exposure burden concentrates.

The 100 Parts-Per-Billion Number

The 100 parts-per-billion figure has, across the regulatory discussion since the Codex 2014 adoption, become the reference number for what comparator jurisdictions consider achievable and protective. The European Union adopted the limit in 2016. The United States Food and Drug Administration set a 100 parts-per-billion final guidance for inorganic arsenic in infant rice cereal in August 2020, finalized as an action level in 2021. China set a 200 parts-per-billion limit for rice in 2017. Australia and New Zealand operate under a 1,000 parts-per-billion limit for inorganic arsenic in rice that has been the subject of ongoing review. The Codex international standard sits at 100 parts per billion for polished rice and 200 for husked.¹²

Healthy Babies Bright Futures, in its May 2025 report, observes that the 100 parts-per-billion limit reflects what regulators consider achievable rather than what the most-protective public-health analysis would recommend. HBBF itself advocates for a 25 parts-per-billion limit, based on its reading of the neurotoxicity literature and the body-weight-adjusted exposure analysis. The campaign's structural argument does not require adjudication of the appropriate limit between 25 and 100 parts per billion. The structural argument is that the absence of any binding limit, in a national regulatory framework, against a documented adoption of the 100 parts-per-billion limit by every comparator jurisdiction, is the regulatory condition under which the disabled household's exposure is being permitted to continue.¹³

The Cumulative Burden

Rice is one source of inorganic arsenic in the contemporary food supply. It is the largest single source for infants, toddlers, and young children, and a disproportionate source for the rice-dependent populations the income framework documented in Chapter I includes in disproportionate numbers. It is not the only source. Apple juice, rice milk, brown rice syrup (used as a sweetener in processed toddler foods and in some infant formulas), rice cakes, rice-based puffed cereals and snack products, and rice-flour-based products in the gluten-free aisle the celiac-diagnosed recipient must purchase from all contribute to the cumulative exposure. The cumulative burden, across the categories, is the figure the longitudinal cohort literature has measured.¹⁴

Healthy Babies Bright Futures' 2019 testing of baby foods across categories — not limited to rice — found that 95 percent of 168 baby foods tested contained at least one heavy metal (arsenic, cadmium, lead, or mercury). The ten most contaminated foods identified in the testing, in descending order of cumulative heavy-metal burden, were: rice cakes; crisped rice cereal; rice-based puffs; brown rice; rice-based teething biscuits and rusks; white rice; raisins; teething crackers (non-rice); granola bar with raisins; and oat-ring cereal. Eight of the ten foods are rice or rice-derived. All ten are typical bottom-cost shelf items the income-constrained household purchases in disproportion to higher-income households.¹⁵

The Constructive Knowledge Doctrine Applied

The constructive knowledge doctrine engaged across the campaign's prior documentation operates against the documented record this chapter assembles. The Government of Alberta is on documented notice — through its own Alberta Health Services Nutrition Guideline; through the Health Canada commitment its provincial counterpart made in 2019 to which Alberta participated in the Codex consultation; through the AISH and ADAP caseload data the same provincial government publishes; and through the Food Banks Alberta HungerCount data on disability-recipient food bank use — that the food supply available to the disabled household at the documented income figure includes the documented exposure burden.

The continued operation of the regulatory framework that permits the exposure on the shelf, alongside the continued operation of the income framework that restricts the disabled household to the bottom-cost shelf, is not an administrative coincidence between two separately optimized systems. It is the operational architecture the chapters that follow document.

What This Chapter Establishes

Four observations follow from this chapter, and the chapters that follow draw on them without restating.

First, inorganic arsenic is present in the rice and rice-based products that constitute the cheapest carbohydrate available on the contemporary Canadian grocery shelf. The presence is documented in 100 percent of 145 rice samples in the most recent primary-source testing. More than one in four of those samples exceeded the 100 parts-per-billion limit adopted by comparator regulatory frameworks.

Second, Health Canada has not adopted a maximum level for inorganic arsenic in rice, in infant rice cereal, or in any rice-based product, despite a 2019 commitment to consult toward a Codex-consistent limit and despite the adoption of the 100 parts-per-billion limit by the European Union (2016), the United States (2021), and the Codex Alimentarius Commission (2014, 2016).

Third, Alberta Health Services has, on the public regulatory record, instructed its own public-health practitioners to advise families that infant rice cereal and rice-based snack products pose a documented arsenic exposure concern. The constructive knowledge condition is met at the provincial level.

Fourth, the household restricted to the bottom-cost grocery shelf by the income framework documented in Chapter I is the household whose consumption of rice-based products is, on the documented household-budget literature, disproportionate. The disabled population the income framework defines is, on the documented record, the population at which the rice-based exposure burden concentrates. The mechanism is not choice. The mechanism is the income framework the same provincial government operates.

Notes — Chapter II

1. European Commission, Commission Regulation (EU) 2015/1006 of 25 June 2015 amending Regulation (EC) No 1881/2006 as regards maximum levels of inorganic arsenic in foodstuffs; U.S. Food and Drug Administration, "Inorganic Arsenic in Rice Cereals for Infants: Action Level," Guidance for Industry, August 2020, finalized 2021; Health Canada, statement of April 4, 2019, on forthcoming consultation toward Codex-consistent maximum levels for inorganic arsenic in rice.

2. Healthy Babies Bright Futures, *What's in My Baby's Food?*, October 2019 (testing of 168 baby foods, 95 percent positive for at least one heavy metal); HBBF, *Arsenic in 9 Brands of Infant Cereal*, 2017 (six-times higher arsenic in infant rice cereal versus other infant cereals).
3. Healthy Babies Bright Futures, *What's in Your Family's Rice?*, released May 15, 2025; hbbf.org/report/arsenic-in-rice. The report tested 145 rice samples purchased at retail stores in 20 metropolitan markets, including U.S.-grown and imported varieties and all major commercial brands distributing into the North American retail market. Food Safety Magazine coverage: "Report: High Levels of Arsenic and Cadmium in Store-Bought Rice Pose Health Threats to Infants," May 20, 2025.
4. Williams, P.N., et al., "Variation in arsenic speciation and concentration in paddy rice related to dietary exposure," *Environmental Science and Technology* 39 (2005): 5531-5540; Meharg, A.A., et al., "Geographical variation in total and inorganic arsenic content of polished (white) rice," *Environmental Science and Technology* 43, no. 5 (2009): 1612-1617. Rice's tendency to accumulate arsenic at approximately ten times the level of other grains is established across the agronomy literature.
5. World Health Organization, IARC Monograph on Arsenic; U.S. Agency for Toxic Substances and Disease Registry, Toxicological Profile for Arsenic, updated 2007 and subsequent revisions; Health Canada, *Guidelines for Canadian Drinking Water Quality: Guideline Technical Document — Arsenic*, May 2006 and subsequent updates. The Group 1 carcinogen classification is established across all major regulatory and scientific bodies.
6. Karagas, M.R., et al., "Association of rice and rice-product consumption with arsenic exposure early in life," *JAMA Pediatrics* 170, no. 6 (2016): 609-616; subsequent publications from the New Hampshire Birth Cohort Study at the Dartmouth Children's Environmental Health and Disease Prevention Research Center, niehs.nih.gov.
7. Body-weight-adjusted exposure is the standard pharmacokinetic adjustment in pediatric environmental health. See, e.g., U.S. Environmental Protection Agency, *Supplemental Guidance for Assessing Susceptibility from Early-Life Exposure to Carcinogens*, EPA/630/R-03/003F, March 2005. The developmental window of central-nervous-system sensitivity is established across the developmental-neurotoxicology literature.
8. Health Canada, *Lists of Maximum Levels for Various Chemical Contaminants in Foods*, canada.ca/en/health-canada/services/food-nutrition/food-safety/chemical-contaminants/maximum-levels-chemical-contaminants-foods.html, accessed May 2026. Codex Alimentarius Commission reference standards, fao.org/fao-who-codexalimentarius/.
9. CBC News, "Health Canada says it will set cap on arsenic in food," April 4, 2019, cbc.ca/news/health/arsenic-rice-baby-food-1.5086078. The CBC Marketplace investigation that precipitated the announcement: CBC Marketplace, "Marketplace tests find arsenic in some Canadian baby food products," March 2019.
10. Codex Alimentarius Commission, General Standard for Contaminants and Toxins in Food and Feed (CXS 193-1995), maximum level for inorganic arsenic in polished rice adopted at the 37th Session, July 2014; maximum level for inorganic arsenic in husked rice adopted at the 39th Session, July 2016. Health Canada has not, as of the date of this edition (May 2026), adopted a corresponding maximum level in Canadian food regulation.
11. Alberta Health Services, Nutrition Services, *Nutrition Guideline — Healthy Infants and Young Children: Arsenic in Foods*, albertahealthservices.ca/assets/info/nutrition/if-nfs-ng-infants-children-arsenic-in-foods.pdf, accessed May 2026. The document is part of the AHS Nutrition Guidelines library and is directed at primary-care practitioners, public-health nurses, and community-health workers.
12. European Commission Regulation (EU) 2015/1006 (June 25, 2015), in force January 1, 2016; U.S. FDA, "Inorganic Arsenic in Rice Cereals for Infants: Action Level," Guidance for Industry, finalized August 2020 (action level) and operationalized 2021. China's GB 2762-2017 sets the inorganic arsenic limit for rice at 200 µg/kg (200 ppb). Australia and New Zealand maximum levels reported in Food Standards Australia New Zealand documentation; the 1 mg/kg (1,000 ppb) limit has been the subject of ongoing review through the joint FSANZ standard-setting process.
13. Healthy Babies Bright Futures, *What's in Your Family's Rice?*, May 2025; HBBF advocates for a 25 parts-per-billion limit, based on the coalition's reading of the body-weight-adjusted exposure literature and the developmental-neurotoxicology evidence base.
14. Healthy Babies Bright Futures cumulative-exposure analysis across the rice-based product category, integrated across multiple HBBF reports from 2017 to 2025; supplemented by the New Hampshire Birth Cohort Study longitudinal exposure data. The contribution of brown rice syrup as a sweetener in toddler foods and in some infant formulas is documented in HBBF, *Arsenic in Infant Formula and Toddler Foods*, 2012.
15. Healthy Babies Bright Futures, *What's in My Baby's Food?*, October 2019. The ten most contaminated foods list is reproduced in HBBF's public materials and in the Healthy Children Project's documentation of the findings (healthychildrenproject.org).

PART TWO · WHAT IS ON THE SHELF**Chapter III****The Colours of Convenience***Synthetic Dyes Permitted in Canada*

Chapter II documented the heavy-metal exposure burden in the carbohydrate category of the bottom-cost grocery shelf. This chapter documents the synthetic-dye exposure burden in the processed-food and shelf-stable-product categories the same shelf is anchored by. The structural argument is the same. The Canadian regulatory framework permits a list of synthetic dyes that comparator regulatory frameworks have, in the documented period 2010 to 2025, progressively required warning labels on, removed from infant and child product lines, or announced phase-out timelines for. The Canadian framework has not done so. The shelf in Canada is, on the regulatory record, less protected than the shelf in comparator jurisdictions.

The List

Health Canada maintains a List of Permitted Colouring Agents under the Food and Drug Regulations, CRC c 870, Division 16. The list authorizes the use in food of the following synthetic colours, identified by their Canadian common name and the United States FDA equivalent name where applicable:¹

- Allura Red AC (FD&C Red No. 40);
- Tartrazine (FD&C Yellow No. 5);
- Sunset Yellow FCF (FD&C Yellow No. 6);
- Brilliant Blue FCF (FD&C Blue No. 1);
- Indigotine (FD&C Blue No. 2);
- Fast Green FCF (FD&C Green No. 3);
- Erythrosine (FD&C Red No. 3);
- Citrus Red No. 2 (restricted to the surface of whole oranges in Canada).

Of the eight U.S. FDA-certified synthetic colours, Health Canada permits all eight. The eighth, Orange B, is not permitted in Canada as a food additive but is permitted in the United States in casings for hot dogs and sausages.²

The European Comparator

The European Food Safety Authority (EFSA) conducted a comprehensive re-evaluation of food colour additives between 2008 and 2014, published as a series of EFSA Scientific Opinions in the EFSA Journal. The re-evaluation drew on a substantial peer-reviewed evidence base, including the 2007 Southampton study published in *The Lancet* (McCann et al., 2007), which documented associations between consumption of a mixture of six synthetic dyes and behavioural effects in children, including hyperactivity and attentional symptoms.³

Effective July 20, 2010, European Regulation (EC) No 1333/2008 requires that any food product containing any of six synthetic food colours bear the warning: "[Name of colour] – may have an adverse effect on activity and attention in children." The six colours requiring the warning are Allura Red AC (E129), Sunset Yellow FCF (E110), Tartrazine (E102), Quinoline Yellow (E104), Carmoisine (E122, banned in Canada and the U.S.), and Ponceau 4R (E124, banned in Canada and the U.S.). The first three of these — Allura Red, Sunset Yellow, Tartrazine — are colours permitted in Canada without any analogous warning-label requirement.⁴

The warning-label requirement is not a ban. It is a notification-disclosure requirement that places the consumer in a position to make an informed purchasing decision. The European framework's conclusion — that consumers are entitled to the documented warning on the product label — has been the operative standard in the European Union for over fifteen years. The corresponding Canadian standard does not include the analogous warning. The Canadian consumer purchasing the same products containing the same dyes receives, on the product label, the dye name and no further information about the documented behavioural-effects literature.⁵

Country-Level Bans

Beyond the European Union's labelling framework, several European countries have imposed national bans on specific synthetic dyes. Allura Red AC, permitted in Canada, is reported to be banned for food use in Denmark, Belgium, France, Germany, Switzerland, Sweden, Austria, and Norway. Several of these national bans pre-date the European Union's 2010 labelling regulation and reflect country-level public-health determinations that the dye is not appropriate for the food supply at all.⁶

The country-level pattern is the documentation of regulatory disagreement among comparator jurisdictions on the appropriate treatment of these dyes. The disagreement is not a contest between the European framework and the rest of the world. It is a contest within the European framework, in which several jurisdictions have determined that the warning-label disclosure does not adequately address the documented concern. The Canadian regulatory framework has not, on the documented record, engaged the contest. It has retained the dyes on the permitted list without the warning-label disclosure and without national-level restriction.

The U.S. April 2025 Announcement

On April 22, 2025, U.S. Health and Human Services Secretary Robert F. Kennedy Jr. and FDA Commissioner Marty Makary jointly announced that the U.S. Food and Drug Administration would phase out the use of petroleum-based synthetic food dyes in the U.S. food supply. The announcement identified eight dyes for action: Citrus Red No. 2, FD&C Green No. 3, FD&C Red No. 40, FD&C Yellow No. 5, FD&C Yellow No. 6, FD&C Blue No. 1, FD&C Blue No. 2, and Orange B. The phase-out timeline announced was "by the end of 2026" for the elimination of the dyes from the U.S. food supply, supported by FDA action to revoke authorizations for the dyes that remain in the regulatory framework.⁷

The U.S. announcement, irrespective of the political context in which it was made and irrespective of the contested public-health framings the announcement was accompanied by, places the United States on the regulatory record as having determined that the petroleum-based synthetic food dyes are not appropriate for the U.S. food supply. The seven dyes identified in the U.S. announcement that Health Canada permits are: Allura Red AC (Red 40), Tartrazine (Yellow 5), Sunset Yellow FCF (Yellow 6), Brilliant Blue FCF (Blue 1), Indigotine (Blue 2), Fast Green FCF (Green 3), and Erythrosine (Red 3, banned by FDA for food use in January 2025 separately from the broader phase-out announcement). All seven remain on the Canadian permitted list as of the date of this edition, May 2026.⁸

Health Canada's Response, April 2025

On April 23, 2025, the day after the U.S. announcement, Health Canada confirmed through written statement to CBC News that the agency had assessed the food colours in question and that, in the agency's determination, "there is no evidence to support changing the rules for these colours in Canada." The agency's position is that the dyes are safe at permitted levels of consumption and that the regulatory framework's requirement for declaration of the colour on the ingredient label provides consumers with the information needed to make purchasing decisions.⁹

Two structural features of Health Canada's April 2025 position bear noting on the documented record. First, the position rests on a safety determination at permitted levels of consumption — the conventional regulatory framework for additives — without engaging the body-weight-adjusted exposure analysis that the European Union, the United States Food and Drug Administration, and the multiple European national bans have engaged. The conventional framework calculates permitted consumption against an adult population at average body weight; the body-weight-adjusted analysis recognizes that infants, toddlers, and young children consume the same products at substantially higher body-weight-adjusted rates and that the developmental window during which the central nervous system is most exposure-sensitive is the same window in which children consume the highest body-weight-adjusted dose. The Health Canada April 2025 position does not, on the public record, engage the body-weight-adjusted analysis.¹⁰

Second, the position rests on the proposition that the ingredient-label disclosure provides consumers with the information they need to make purchasing decisions. The proposition assumes that the consumer reading the label has the regulatory literacy to recognize that "Allura Red" is Red 40, that Red 40 is among the dyes the European Union requires a hyperactivity warning on, and that the behavioural-effects literature exists and would inform the purchasing decision. The disability cohort the income framework defines — 28.3 percent with primary cognitive presentations, 30.2 percent with primary mental-illness presentations — is the cohort least likely, on the documented cognitive-access literature, to perform the regulatory-literacy task the Health

Canada position assumes. The campaign's *Access Gap: Penalized for Non-Ability* documents the broader pattern of which the food-label literacy question is a sub-instance.¹¹

The Manufacturer-Discretion Gap

The Canadian regulatory framework's permitting of the dyes creates space for a documented manufacturer-discretion pattern. The same brand-name product, sold in the United States and in Canada, is in many cases manufactured to different formulations for the two markets. The example most commonly cited in the public discussion is Kellogg's Froot Loops, manufactured with synthetic petroleum-based dyes in the United States and with fruit-based and vegetable-based natural colourings in Canada. The Canadian Froot Loops formulation is the documented evidence that the alternative formulation is commercially feasible. The Kellogg decision to use the alternative formulation in Canada is, on the documented record, a market choice the company has made; it is not a regulatory requirement.¹²

The manufacturer-discretion gap is the documented condition under which two things follow at the level of the bottom-cost grocery shelf. First, the consumer purchasing the brand-name product in Canada cannot reliably know, from the product label alone, whether the formulation in the package is the same formulation the consumer would receive in the United States or in the European Union. The ingredient label discloses the dyes used in the Canadian formulation of the specific product; it does not disclose whether the same product is differently formulated elsewhere. Second, the cheapest products on the bottom-cost grocery shelf are typically not the brand-name products that Kellogg or comparable major manufacturers have chosen to reformulate for the Canadian market. They are the lower-priced private-label products and the lower-priced second-tier brand products in which the manufacturer has, on the documented public record, made the different commercial decision to retain the synthetic dyes at the price-point the income-constrained consumer accesses.¹³

The Permitted Additives Beyond the Dyes

This chapter has focused on the synthetic dye category for the directness of the comparator-jurisdiction record. The broader category of food additives permitted in Canada and restricted, labelled, or banned in comparator jurisdictions is larger. Sodium benzoate, a preservative used in pickled and acidic shelf-stable products, is permitted in Canada and is included in the European Union's hyperactivity-warning regulatory framework when combined with the regulated dyes. Butylated hydroxyanisole (BHA) and butylated hydroxytoluene (BHT), used as preservatives in shelf-stable cereal, gum, and processed-meat products, are permitted in Canada and have been restricted or required to be labelled in several European jurisdictions. Potassium bromate, used as a flour treatment in the bread-baking industry, was banned in Canada for food use in 1994; the Canadian framework on this specific additive is the protective framework, and is identified here in the interests of accuracy and balance.¹⁴

The broader-additives picture confirms the directional pattern this chapter documents. The Canadian regulatory framework is not uniformly more permissive than the European Union or United States frameworks; in some additive cases it has acted earlier and more restrictively. The pattern of relative permissiveness is, however, the documented pattern across the synthetic-dye category and the category most directly relevant to the bottom-cost shelf the income-constrained household reaches.

What This Chapter Establishes

Four observations follow from this chapter.

First, the Canadian regulatory framework permits the use in food of seven synthetic dyes that the U.S. Food and Drug Administration announced a phase-out of in April 2025, and that the European Union has required a hyperactivity-effects warning label on (or has restricted further at the national level in several jurisdictions) since 2010.

Second, Health Canada's April 2025 position, in response to the U.S. announcement, is that there is no evidence supporting a change in Canadian rules. The position rests on a conventional safety-at-permitted-consumption framework that does not engage the body-weight-adjusted exposure analysis comparator jurisdictions have engaged, and on an ingredient-label disclosure framework that assumes a regulatory literacy the disability cohort is, on the documented record, least likely to have.

Third, the manufacturer-discretion gap permits the same brand-name product to be differently formulated for the Canadian and the U.S. or European markets. The consumer purchasing the cheapest products on the bottom-cost shelf does not, in most cases, receive the reformulated brand-name product; the reformulation has been commercially confined to the higher-price-point brand-name segments. The lower-price-point segments — private label, second-tier brands — retain the synthetic dyes the U.S. framework is phasing out and the European framework requires warning labels on.

Fourth, the structural condition this chapter and Chapter II together document is the condition under which the disabled household, restricted by the income framework to the bottom-cost shelf, purchases products containing exposures the Canadian regulatory framework permits and that comparator regulatory frameworks have determined require labelling, restriction, or phase-out. The chapters that follow document the population-level consequence of the shelf condition this chapter and the preceding chapter have established.

Notes — Chapter III

1. Food and Drug Regulations, CRC c 870, Division 16, sections C.01.040.2 and subsequent; Health Canada, *List of Permitted Food Colours* (Lists of Permitted Food Additives), canada.ca/en/health-canada/services/food-nutrition/food-safety/food-additives/lists-permitted/3-colouring-agents.html, accessed May 2026.
2. Health Canada, written statement to CBC News, reported in Yoon, J. and Zafar, A., "RFK Jr. calls artificial food dyes 'poisonous.' Here's how they're regulated in Canada," CBC News, April 23, 2025, cbc.ca/news/health/food-dye-1.7516794.
3. McCann, D., et al., "Food additives and hyperactive behaviour in 3-year-old and 8/9-year-old children in the community: a randomised, double-blinded, placebo-controlled trial," *The Lancet* 370, no. 9598 (3 November 2007): 1560-1567, doi:10.1016/S0140-6736(07)61306-3. The European Food Safety Authority Scientific Opinions on the re-evaluation of the relevant colours are published in the EFSA Journal, 2008-2014.
4. European Parliament and Council Regulation (EC) No 1333/2008 of 16 December 2008 on food additives, Annex V, requiring the specified warning on foods containing the listed colours; in force July 20, 2010. The six colours subject to the warning requirement are E129 (Allura Red), E110 (Sunset Yellow), E102 (Tartrazine), E104 (Quinoline Yellow), E122 (Carmoisine), and E124 (Ponceau 4R).
5. The European Union labelling framework remains in force as of the date of this edition (May 2026). The Canadian Food and Drug Regulations, CRC c 870, do not contain an analogous warning-label requirement.

6. Country-level bans on Allura Red reported across multiple sources including the Council of Canadians' analysis "Food dye warning labels and CETA" (canadians.org/analysis/news-food-dye-warning-labels-and-ceta/) and CBC News reporting through 2022-2025. The specific status of each national ban should be verified against the relevant national-level regulatory source.
7. U.S. Department of Health and Human Services and U.S. Food and Drug Administration, joint announcement, April 22, 2025; coverage: Yoon, J. and Zafar, A., CBC News, April 23, 2025, op. cit.; multiple contemporaneous U.S. media accounts.
8. FDA revocation of authorization for FD&C Red No. 3 for food use, announced January 15, 2025, with manufacturers required to reformulate by January 15, 2027. The action on Red 3 preceded the broader April 22, 2025 announcement on the remaining petroleum-based synthetic dyes. Health Canada has not announced corresponding action on Erythrosine (Red 3) for food use; the colour remains on the Canadian permitted list.
9. Health Canada written statement to CBC News, April 23, 2025, reported in Yoon and Zafar, CBC News, op. cit.
10. Body-weight-adjusted exposure analysis in pediatric environmental health is documented in U.S. EPA, *Supplemental Guidance for Assessing Susceptibility from Early-Life Exposure to Carcinogens*, EPA/630/R-03/003F, March 2005, and in the broader pediatric-pharmacokinetics literature.
11. AISH caseload composition data: Government of Alberta, AISH Caseload Open Data, September 2025 (79,290 recipients; 22,419 cognitive (28.3 percent); 23,940 mental illness (30.2 percent); 46,359 combined cognitive-and-mental-illness (58.5 percent)). The Alberta Disability System Breakdown, *The Access Gap: Penalized for Non-Ability*, May 2026, on the broader regulatory-literacy and cognitive-access pattern.
12. Kellogg's Froot Loops formulation comparison documented in multiple consumer publications and trade press; the Canadian formulation uses fruit and vegetable juices (blueberry juice, carrot juice, watermelon juice concentrate) where the U.S. formulation uses FD&C Yellow No. 6, FD&C Red No. 40, and FD&C Blue No. 1. The Kellogg corporate decision is documented but is not a regulatory requirement.
13. The manufacturer-discretion pattern across the Canadian market is documented in trade publications and in the price-point analysis available in the Statistics Canada Consumer Price Index data on the processed-food and breakfast-cereal category. The pattern of reformulation being confined to the higher-price-point brand-name segment is the documented pattern in the bottom-cost-shelf consumer research.
14. Sodium benzoate (E211), BHA (E320), and BHT (E321) are documented in the European Union additives framework and in the Canadian Food and Drug Regulations permitted list. Potassium bromate (E924) was prohibited in Canada in 1994 by amendment to the Food and Drug Regulations, several years in advance of the European Union prohibition. The Canadian framework is the protective framework on the potassium bromate question; the chapter retains the observation to maintain the balance of the documented record.

PART THREE · THE STRUCTURAL CONCENTRATION

Chapter IV

The Food Bank as Symptom

Disability and Hunger in Alberta

Chapters I, II, and III documented the structural condition at the household level: the income framework restricts the disabled household to the bottom-cost grocery shelf; the bottom-cost shelf contains the documented exposure burden the Canadian regulatory framework permits and that comparator frameworks have restricted. This chapter documents what happens when the income framework restricts the household below the point at which the bottom-cost shelf can be afforded at all. The food bank, in the documented Alberta data, is the structural outcome.

The argument the chapter develops draws on the most recent primary-source analysis: the May 22, 2026 report from the University of Calgary's School of Public Policy, lead-authored by Dr. Ron Kneebone, that uses Calgary Food Bank operational data to document the food-bank-use patterns of households on Alberta disability income assistance. The report's findings are released into a national context — the Food Banks Canada 2024 HungerCount — in which Alberta's food-bank growth since 2019 is the largest in Canada at 134.4 percent.

The Kneebone Findings, May 22, 2026

The University of Calgary School of Public Policy report, published May 22, 2026, uses anonymized Calgary Food Bank client-level data to analyze the duration and intensity of "food-bank episodes" by client-level characteristics including primary income source. The report's central finding, as summarized in the Medicine Hat News coverage on the publication date, is that "households relying on the Assured Income for the Severely Handicapped program rely on food banks to offset grocery costs more than other households." The food-bank episode — defined in the methodology as the duration across which a household accesses the food bank with documented regularity — is, for AISH households, longer and more frequent than for other client populations.¹

Dr. Kneebone's framing of the finding is methodological as well as substantive. Food bank data, he argues in the report, is "a valuable and largely untapped resource governments can use to measure the success of income support programs." A previous Kneebone report had documented food bank use as an indicator for households at risk of first-time homelessness — the food bank visits escalating in the five years preceding a first homeless-shelter use. The same methodological framework applied to the AISH cohort places food-bank use as the structural indicator that the income figure is, against the cost of living, insufficient to sustain the household. The report's framing is not that the AISH cohort makes different consumption choices than other cohorts. It is that the income figure structurally produces the food-bank-dependence pattern the data document.²

The Medicine Hat News coverage of the report cites a 6.4 percent share of Alberta food bank users whose primary income source is provincial disability support, against a national average of 11.7 percent for the same category. The Alberta figure appears lower than the national figure on first reading; the report's analytic point is that the Alberta number understates the structural condition. Approximately 1.6 percent of Alberta's population is on AISH; the 6.4 percent share of food bank users is four times the population share. The disproportionality is the indicator the Kneebone analysis is built on, and the Alberta number is, on the population-adjusted basis, evidence of the same structural disproportionality the national figure represents.³

The Provincial-Disability Share Across Time

The Food Banks Canada HungerCount series provides the longitudinal dataset against which the Kneebone findings sit. The 2021 HungerCount, reflecting March 2021 client-level data across Alberta food banks, reported that 15.2 percent of Alberta food bank users received Provincial Disability Supports as a primary income source. The same report identified that in rural Alberta, 30.5 percent of food bank users were seniors or persons with a disability. The 2021 HungerCount recorded a 29.6 percent increase in food bank access in Alberta against the 2019 HungerCount baseline.⁴

The 2024 HungerCount, the most recent national release with detailed disability-cohort breakdowns, identified provincial disability support as the primary income source for 12.2 percent of Canadian food bank clients nationally in 2024 — down from 17.3 percent in 2019. The shift in shares is not a reduction in absolute disability food-bank use; it is the effect of substantial growth in the other client categories (employed clients, government-assistance clients) during the post-pandemic period, against a baseline of disability-cohort use that has remained at or above its pre-pandemic level. The denominator has grown faster than the disability share has grown. The disability cohort's food-bank use, in absolute terms, has continued to rise.⁵

The Alberta-specific 2024 HungerCount data, published in the Food Banks Canada Alberta 2024 Report Card, documents an overall food bank visit total of 172,832 for Alberta in 2024, including 61,379 child visits (35 percent of all visits). The same report records that the Alberta increase in food bank visits between 2019 and 2024 was 92.4 percent — and that the increase to March 2025 reached 134.4 percent — the largest provincial increase in Canada across the same period.⁶

The 2025 Trajectory

The trajectory of food-bank use in Alberta from 2024 to 2025 is the trajectory the Kneebone report is documented against. Food Banks Canada's national figures released in May 2025 indicate that food bank visits across Canada grew by more than five percent year over year to March 2025; in Alberta, the year-over-year growth was 21.8 percent. The cumulative growth in Alberta food bank visits from March 2019 to March 2025 reached 134.4 percent. The Alberta growth is the largest provincial-level growth in the country across the same comparison period.⁷

The trajectory matters to the structural argument because the trajectory is running in parallel with two policy developments the campaign's prior documentation has tracked. Bill 12, passed in December 2025, removed the legislated indexation mechanism that adjusted AISH benefits against inflation. The ADAP regulation filed in May 2026, AR 96/2026, in force July 2, 2026, transitions approximately 50,000 current AISH recipients to the lower ADAP rate. Both policy changes will operationalize, in the period after this edition goes to print, against an Alberta food-bank-use trajectory already moving in the documented direction.

The structural argument is not that the policy changes will produce a food-bank-use spike in some future period. The structural argument is that the trajectory is already in motion and the policy changes will accelerate it.⁸

The Income Support Surge

Alongside the food-bank trajectory, the Alberta Income Support caseload — a separate provincial program from AISH but with substantial overlap in cohort characteristics — has documented its own substantial increase. CBC News reported in January 2025 that the Income Support caseload had surged to approximately 57,000 in 2024, with Maytree's Social Assistance Summaries data confirming 57,347 Income Support cases against 77,647 AISH cases in the 2024-25 fiscal year. The total provincial social-assistance caseload, including both programs, reached approximately 134,994 family units and unattached single adults — the largest provincial social-assistance caseload Alberta has carried since the pre-pandemic period.⁹

The Maytree analysis identifies that 4.3 percent of Alberta's population under 65 received Income Support or AISH in 2024-25 — approximately one in 23. The Income Support program — which is the program operative for non-disabled poor Albertans whose situations do not qualify them for AISH — is, against the Maytree framework, the program providing the least adequate support in Canada: less than 40 percent of the poverty line for a single person living in Calgary. The disability cohort the AISH program serves sits at a higher absolute income figure but at an income figure that is, against the documented cost of living and the documented household-budget arithmetic, still substantially below the poverty line.¹⁰

The Calgary Food Bank Operational Data

The Kneebone analysis draws on what is, in the Canadian food-bank-research community, the most operationally robust client-level dataset: the Calgary Food Bank's intake and visit records, retained at the household-identifier level across multi-year periods. The Calgary Food Bank's own published statistics document, for fiscal year 2024, approximately 172,832 visits attributable to Alberta food bank locations across the province, of which the Calgary-area share is approximately one-third. Sixteen percent of food bank recipients in 2024 are fully employed; the majority of the remainder are on income support, on AISH, or in transition between income sources.¹¹

The longitudinal pattern in the Calgary Food Bank data — the pattern the Kneebone analysis treats as its analytical foundation — is that the disabled household, once it enters the food-bank-episode pattern, exits it more slowly than other client populations and re-enters it more frequently. The structural interpretation of the pattern is that the income framework that determined the household's entry into the pattern does not, on the documented data, contain a mechanism by which the household can be moved out of it. The food bank, in the Kneebone framing, is not a transitional support that the household passes through. It is, for the AISH and disability-supported cohort, a structural part of the household budget.¹²

The Implications for the Food Supply Analysis

The food bank, in the structural argument this piece develops, is the resolution of the impossibility documented in Chapter I. The household whose residual after rent and the documented non-food fixed expenses is below the food cost the Statistics Canada figure establishes cannot purchase food at the documented Canadian grocery price point. The household resolves the impossibility through some combination of three mechanisms: cuts to the other expense categories (documented in Chapter I as the medication-non-adherence pattern, the missed-medical-appointment pattern, and the cut-heat-in-winter pattern); reliance on the food bank; and reliance on familial or community subsidy from members of the household's network who are themselves not on a disability benefit.¹³

The food bank, considered as a source of the household's monthly food supply, operates within its own documented constraints. Food banks distribute, in substantial proportion, the same shelf-stable processed-food category Chapter III documents — donated by manufacturers, supermarket chains, and food drives from across the broader Canadian food system. The food bank's role is to provide nutrition to households whose income does not allow them to purchase food at the documented price points. The food bank's role is not to provide the population it serves with access to the higher-quality shelf segment. The household that enters the food bank is, in the documented operational pattern, drawing from a food supply with the same general composition as the bottom-cost grocery shelf — and with the same documented exposure burden.¹⁴

The argument is not that food banks are doing harm to the populations they serve. Food banks are providing emergency food relief to populations whose income does not allow them to purchase food. The food banks operate within the food supply that the Canadian regulatory framework permits and that the Canadian manufacturing system produces. The structural concern the chapter documents is that the income framework's progression — Bill 12's elimination of indexation; the AISH-to-ADAP transition's reduction of the benefit level; the Civida community-housing rent recalibration's compounding effect on the household's residual; the CDB clawback's nullification of the federal poverty-reduction intervention — moves more disabled households into the food-bank-dependence pattern, and that the food supply the food bank distributes is, on the documented record, the supply with the documented exposure burden.¹⁵

What This Chapter Establishes

Four observations close this chapter.

First, the food-bank-use pattern in Alberta has grown by 134.4 percent in the period from March 2019 to March 2025, the largest provincial growth in Canada. The Alberta growth has continued at 21.8 percent year-over-year into 2025 — the period in which Bill 12 was enacted, the ADAP regulation was filed, and the AISH-to-ADAP transition began its operational implementation phase.

Second, the May 22, 2026 University of Calgary School of Public Policy report by Dr. Ron Kneebone documents that AISH households use food banks more frequently and for longer episodes than other client populations. The report's methodological framing — food-bank data as the structural indicator of income-framework inadequacy — places the disability cohort's food-bank-use pattern as the documented evidence of the income framework's operational failure to sustain the population it serves.

Third, the food-bank-use pattern is not a transitional support the disabled household passes through. In the documented longitudinal data, it is a structural part of the household budget. The household that enters the food-bank-episode pattern exits it more slowly than other populations and re-enters it more frequently. The pattern is the documented operational feature of the income framework, not a behavioural outcome at the household level.

Fourth, the food supply the food bank distributes draws from the same Canadian food-manufacturing system that supplies the bottom-cost grocery shelf. The documented exposure burden Chapter II and Chapter III have established does not exit the food supply at the food bank's door. The household whose income moves it from the bottom-cost grocery shelf to the food bank is moving across the income line, not across the exposure line. The exposure follows the food supply, not the recipient's program-eligibility category.

Notes — Chapter IV

1. Medicine Hat News, "Report says Albertans on disability supports disproportionately use food banks," May 22, 2026, medicinehatnews.com/news/local-news/2026/05/22/, reporting on the University of Calgary School of Public Policy report lead-authored by Dr. Ron Kneebone, released the same day.
2. Statement of Dr. Ron Kneebone, scientific director of social policy and health research, University of Calgary School of Public Policy, reported in Medicine Hat News op. cit.; previous Kneebone work on food bank use as a homelessness indicator referenced in the same coverage.
3. Medicine Hat News, May 22, 2026, op. cit. The 6.4 percent Alberta-specific share of food bank users on provincial disability supports compared to the 11.7 percent national average; the 1.6 percent Alberta-population share on AISH; the four-times-population-share disproportionality is the campaign's analytical observation drawn from the published figures.
4. Food Banks Alberta (Arianna Scott, Interim CEO), 2021 HungerCount Findings for Alberta, November 2021, reproduced in Prairie Post (Lethbridge), November 5, 2021. The 15.2 percent of Alberta food bank users on Provincial Disability Supports and the 30.5 percent rural figure for seniors and persons with a disability are the published Food Banks Alberta figures.
5. Food Banks Canada, HungerCount 2024 national figures; The Hub, "Nearly one in five Canadians using food banks have jobs," November 15, 2024 (reporting and analysis of the 2024 HungerCount client-source breakdowns showing the relative-share shift across categories).
6. Food Banks Canada, Alberta 2024 Report Card, October 28, 2024, foodbankscanada.ca/poverty-report-card/report-card/?p=alberta&y=2024; Bluesky Calgary (data-driven analysis of Calgary Food Bank statistics), April 2, 2025; figures of 172,832 total Alberta food bank visits, 61,379 child visits (35 percent of all visits), and the 2019-to-2024 visit increase of 92.4 percent.
7. Food Banks Canada, March 2025 HungerCount national data; Alberta year-over-year increase of 21.8 percent; cumulative 2019-to-2025 increase of 134.4 percent — largest provincial increase in Canada across the same period. Coverage: Medicine Hat News, May 22, 2026, op. cit.
8. Bill 12, Financial Statutes Amendment Act, 2025 (No. 2), SA 2025 c20, removed the legislated AISH indexation mechanism. AR 96/2026, the ADAP regulation, in force July 2, 2026, transitions approximately 50,000 current AISH recipients to the lower ADAP rate. The Civida community-housing rent recalibration applies a 30-percent-of-household-income rent formula at annual income reviews; Canada Disability Benefit Act, SC 2023, c 9; the Alberta clawback documented in *The Alberta Disability System Breakdown*, AISH Evidentiary Report Series (April 2026).
9. CBC News, "Number of Albertans relying on income support surges to 57,000," January 6, 2025, cbc.ca/news/canada/edmonton/alta-income-support-1.7423736; Maytree Foundation, Social Assistance Summaries, Alberta, March 31, 2026 update, maytree.com/changing-systems/data-measuring/social-assistance-summaries/alberta/. The 2024-25 totals: 57,347 Income Support cases; 77,647 AISH cases; 134,994 total cases.
10. Maytree Social Assistance Summaries, Alberta, op. cit.; Food Banks Canada, Alberta 2024 Report Card, op. cit. (Income Support rates as share of poverty line for a single adult in Calgary).

11. Calgary Food Bank, published statistics, FY 2024; Bluesky Calgary, "A Data-Driven Look at Calgary's Food Bank Statistics [2026]," April 2, 2025, blueskycalgary.com/food-bank-statistics-calgary/.
12. Kneebone analysis of Calgary Food Bank household-level longitudinal data, May 22, 2026; the analytical observation that the AISH cohort exits the food-bank-episode pattern more slowly and re-enters more frequently is reported in the Medicine Hat News coverage and is consistent with the broader Kneebone research program on food bank use as a structural indicator.
13. The three-mechanism resolution framework (cuts to other expense categories, reliance on the food bank, reliance on familial or community subsidy) is documented in the broader Canadian food-insecurity literature; see, e.g., Statistics Canada Canadian Income Survey food-insecurity module data and the PROOF Food Insecurity Policy Research program at the University of Toronto (proof.utoronto.ca).
14. The composition of food-bank donated supply is documented across the Food Banks Canada and Food Banks Alberta operational reports; the predominant shelf-stable processed-food share reflects donation patterns from the broader Canadian food-manufacturing and grocery-retail system.
15. The compounding effect of Bill 12, AR 96/2026, the Cvida rent recalibration, and the CDB clawback on the household budget is documented in *The Alberta Disability System Breakdown, Income Crossover Analysis* (revised May 2026); *Compound Failure Brief* (April 2026); and *The Cage Doesn't End* compilation (May 2026).

PART THREE · THE STRUCTURAL CONCENTRATION

Chapter V

The Compound Effect

Exposure Burden Concentrated in the Disability Cohort

The chapters that precede this one have documented four separate elements. The income framework restricts the disabled household to the bottom-cost grocery shelf. The bottom-cost shelf contains documented exposure burden across the heavy-metal category and the synthetic-dye category. The Canadian regulatory framework permits the exposure that comparator frameworks have restricted, labelled, or announced phase-out timelines for. The food bank, which receives the household whose income falls below the bottom-cost-shelf threshold, distributes a food supply with the same composition as the shelf.

This chapter integrates the four elements. The claim it develops is structural and population-level. The disabled population the AISH and ADAP framework defines is the population at which the exposure burden documented in the preceding chapters concentrates. The concentration is not a side effect of the income framework, the food regulatory framework, the food-manufacturing system, or the food-bank-distribution architecture. It is the cumulative operational outcome of all four operating together, on the same population, across the same period.

The Cohort Composition

The AISH caseload as of September 2025, the most recent published data, comprises approximately 79,290 individuals. The Government of Alberta's open data identifies primary medical condition across the caseload as follows: 22,419 (28.3 percent) with cognitive disorders as primary diagnosis, including autism, intellectual disability, acquired brain injury, and severe dementia; 23,940 (30.2 percent) with mental illness as primary diagnosis, including major depressive disorder, severe anxiety, schizophrenia, and post-traumatic stress disorder; approximately 24,800 (31.3 percent) with physical disability, including mobility, sensory, and chronic-illness conditions; and approximately 8,131 (10.2 percent) with other or multiple primary diagnoses. Combined, 46,359 people — 58.5 percent of the AISH caseload — have a primary diagnosis of cognitive disorder or mental illness.¹

The cohort composition matters to this chapter's argument for three reasons. First, the conditions present in the majority of the caseload are conditions whose presentation overlaps, at the level of clinical symptomatology and neurodevelopmental history, with the documented outcomes the exposure literature associates with the contaminants the preceding chapters have documented. The argument the chapter develops is not that the food-supply exposures caused these conditions. The argument is that the cohort already living with the conditions is being concentrated, by the income framework, in the food-supply segment with the documented exposure burden. Whether the exposure burden contributes to symptom exacerbation, to

medication-effectiveness reduction, to cognitive-functioning fluctuation, or to none of these, the cohort's exposure is being structurally elevated above the population average.²

Second, the AISH caseload includes children. Approximately twelve percent of the caseload — close to 9,500 individuals — are aged 18 to 25, the cohort whose transition out of the FSCD framework into the adult AISH framework is documented in the campaign's prior reports. The children of AISH recipients are not in the caseload number, but they are in the household. The Government of Alberta has not published, in any disaggregated form, a count of children in AISH households; the Inclusion Alberta survey work and the broader Alberta family-composition statistics suggest that approximately 20 to 25 percent of AISH households include dependent children. Applied to the 79,290 caseload, the implied estimate is approximately 16,000 to 20,000 children in AISH households as of September 2025.³

Third, the cognitive and mental-illness presentations that dominate the caseload are precisely the presentations whose downstream effects include the documented difficulty navigating bureaucratic systems, the documented difficulty reading and processing complex written communication, and the documented difficulty with the multi-step administrative tasks that the food-labelling literacy Chapter III examined assumes. The cohort least able, on the documented record, to read the ingredient label, to recognize that "Allura Red" is the same substance as "Red 40," to find and apply the European Food Safety Authority Scientific Opinion to a purchasing decision, is the cohort whose income is structurally concentrated against the products whose labels would, in a comparator regulatory framework, carry the warning.⁴

The Developmental Window

The exposure literature documented in Chapter II identifies a specific developmental window during which the central nervous system is most sensitive to the neurotoxic effects of inorganic arsenic, lead, cadmium, and the broader heavy-metal category. The window extends from conception through approximately age six, with peak sensitivity in the period from gestation through age two. The same window is the window during which children of AISH and ADAP households are being raised on the bottom-cost food supply Chapter II documented. The body-weight-adjusted exposure is the figure that matters: an infant or toddler consumes rice and rice-based products at a body-weight-adjusted rate approximately three times higher than an adult, and the cumulative exposure across the window is the figure the longitudinal cohort literature has measured.⁵

The FSCD population — children of any household whose disability is severe enough to qualify for Family Support for Children with Disabilities — is a population that includes children of AISH households and children of non-AISH households. The cohort overlap is structurally relevant: a child diagnosed with a developmental disability at age three whose parent is an AISH recipient is a child whose disability is already qualifying for the provincial FSCD framework, whose family income is at the AISH level documented in Chapter I, whose family's food budget is at the residual level documented across the preceding chapters, and whose body-weight-adjusted exposure during the developmental window is the figure this chapter is examining.⁶

The campaign's documentation has not, on the available data, undertaken a primary-source analysis of the body-weight-adjusted exposure of children in FSCD-eligible families on AISH-level income. The analysis would require primary-source food-frequency data the campaign does not have access to. The structural observation this chapter makes is more modest: that the conditions for elevated exposure in the developmental window — income restriction to the bottom-cost shelf, household reliance on the rice-based and shelf-stable

categories, and the absence of the protective Canadian regulatory framework that would apply to the products on the shelf — are documented at the household level. The exposure-elevation pathway is operationally available. The epidemiological measurement of the population-level outcome is the work the Dartmouth Children's Environmental Health and Disease Prevention Research Center and comparable research programs have begun, and have not yet completed for the Canadian disability cohort specifically.

The Constructive Knowledge Condition

The constructive knowledge doctrine engaged across the campaign's documentation operates against the documented record this chapter integrates. The Government of Alberta is on documented notice — through its own Alberta Health Services Nutrition Guideline on Arsenic in Foods (Chapter II); through its participation in the federal-provincial-territorial Codex consultation Health Canada announced in 2019 (Chapter II); through its own AISH and ADAP caseload composition data (this chapter); through the Food Banks Alberta and Food Banks Canada HungerCount surveillance data the same provincial government receives copies of (Chapter IV); through the Kneebone May 22, 2026 University of Calgary School of Public Policy report distributed across the Alberta public-policy and political-press community (Chapter IV) — that the cumulative condition this chapter documents is the cumulative condition this chapter documents.⁷

The continued operation of the four documented elements — the income framework at the AISH and ADAP levels; the absence of a Canadian regulatory limit for inorganic arsenic in the rice-based category; the Canadian permitting of the synthetic dyes the U.S. and EU frameworks have moved against; the food-bank distribution architecture that operates within the same food-manufacturing system — is not, on the constructive knowledge doctrine, a series of separately optimized administrative choices that happen to coincide on the same population. It is the operating architecture under which the documented exposure burden has been allowed to concentrate on the cohort the income framework defines.

The structural argument does not require any individual policy designer, food regulator, food manufacturer, food-bank director, or government employee to have intended the cumulative outcome. The argument is institutional and cumulative. The institution that operates the income framework is on notice of the food cost. The institution that operates the food-safety framework is on notice of the comparator regulatory record. The institution that operates the AISH and ADAP programs is on notice of the cohort composition. The cumulative operational outcome — that the disability cohort the income framework defines is the cohort at which the documented exposure burden concentrates — is the documented institutional outcome of the four operating frameworks running in parallel.⁸

The Architecture of the Inheritance

The architecture this chapter describes is, in its structure, the same architecture the campaign's prior documentation has named. *The Cage Doesn't End* identified the structural condition under which the disabled household is restricted, by the integrated effect of income, cohabitation, partnership, and benefit-clawback rules, to the conditions of a life that the broader Canadian population does not face. *Consent Was Never the Point* identified the historical continuity between the explicit eugenic framework of 1928-1972 and the contemporary frameworks that have, through different mechanisms, produced structurally continuous outcomes for structurally continuous populations. *The Access Gap: Penalized for Non-Ability* identified the operational pattern in which the same population the disability framework is designed to serve is the population the framework's compliance requirements are structurally inaccessible to.⁹

The food supply is the fourth element. The disabled household is, in the documented integrated architecture, restricted in income, restricted in partnership and household composition, restricted in administrative access to the supports the disability framework is presented as providing, and — as this piece documents — restricted in the food-supply segment from which it can purchase the household's nutrition. The four restrictions operate together. The food-supply restriction is the restriction this piece adds to the documented record.

The historical parallel is the parallel the campaign's documentation has drawn across its prior reports. The 1928-1972 eugenic framework operated through the explicit mechanism of sterilization authority. The contemporary frameworks operate through the cumulative mechanism of income restriction, regulatory permissiveness, and administrative inaccessibility. The mechanism has changed. The architecture — that some populations are structurally consigned to the conditions of a life the broader population does not face — has not.¹⁰

What This Chapter Establishes

Five observations close this chapter.

First, the disability cohort the AISH and ADAP framework defines is the cohort at which the documented exposure burden — heavy metals in the rice-based category; synthetic dyes in the shelf-stable processed-food category; the broader additive load of the bottom-cost grocery shelf — concentrates. The concentration is the cumulative operational outcome of the income framework, the food-regulatory framework, the food-manufacturing system, and the food-bank-distribution architecture running in parallel on the same population.

Second, the children of AISH households — estimated at approximately 16,000 to 20,000 across the 79,290 caseload — are being raised on the bottom-cost food supply during the developmental window the exposure literature identifies as the period of peak central-nervous-system sensitivity. The body-weight-adjusted exposure during the window is the figure the longitudinal cohort literature has documented as the figure with the largest measurable downstream effects.

Third, the cohort that is, on the documented record, least able to perform the regulatory-literacy task the Canadian ingredient-label disclosure framework assumes is the cohort most concentrated against the products whose labels would, in a comparator regulatory framework, carry the documented warning. The

regulatory-literacy assumption operates as a documented exclusion of the cohort the protective framework would otherwise serve.

Fourth, the constructive knowledge doctrine is engaged on the documented record. The Government of Alberta is on documented notice — through its own Alberta Health Services Nutrition Guideline; through the federal-provincial-territorial Codex consultation record; through its own AISH and ADAP caseload composition data; through the Food Banks Alberta and Food Banks Canada HungerCount surveillance; through the Kneebone May 22, 2026 University of Calgary report — of every element this chapter documents. The continued operation of the integrated architecture, in the face of the documented constructive knowledge, is the operational architecture under which the exposure burden has been allowed to concentrate on the cohort the income framework defines.

Fifth, the architecture this chapter describes is structurally continuous with the architecture the campaign's prior documentation has named across its other reports. The food-supply element is the element this piece adds to the documented record. The closing chapter that follows examines the question of what the documented record this piece has assembled is the documented record of, and what would be required, on the documented record, to alter the architecture.

Notes — Chapter V

1. Government of Alberta, AISH Caseload Open Data, September 2025, open.alberta.ca. Primary medical condition breakdown: 22,419 cognitive (28.3 percent); 23,940 mental illness (30.2 percent); approximately 24,800 physical (31.3 percent); approximately 8,131 other or multiple (10.2 percent). Combined cognitive-and-mental-illness: 46,359 (58.5 percent).
2. The literature on heavy-metal exposure and pre-existing neurodevelopmental and mental-health conditions is documented across the peer-reviewed environmental health and pediatric medicine literature. The campaign does not assert population-level causation between food-supply exposure and the conditions present in the AISH caseload; the argument is structural and population-level concentration.
3. AISH caseload age-distribution data from Government of Alberta AISH open data; the 18-to-25 cohort estimate of approximately twelve percent is consistent with the published caseload composition. The estimate of approximately 20-to-25 percent of AISH households including dependent children draws on Inclusion Alberta survey data and Statistics Canada family-composition statistics for low-income Alberta households. The Government of Alberta has not, on the public record, published a disaggregated count of dependent children in AISH households.
4. The campaign's *The Access Gap: Penalized for Non-Ability* (May 2026) documents the broader pattern of which the food-label literacy observation is a sub-instance. The pattern of the cohort least able to perform the regulatory-literacy task being the cohort the framework structurally requires to perform it is documented across the campaign's evidentiary report series.
5. Developmental window of central-nervous-system sensitivity documented across the developmental-neurotoxicology literature; New Hampshire Birth Cohort Study longitudinal data on infant arsenic exposure at the Dartmouth Children's Environmental Health and Disease Prevention Research Center; U.S. EPA *Supplemental Guidance* on early-life carcinogen susceptibility (EPA/630/R-03/003F, 2005); see Chapter II of this piece for the detailed treatment.
6. Government of Alberta, Family Support for Children with Disabilities (FSCD) program documentation; Inclusion Alberta FSCD Survey Report (January 2025); cohort overlap between FSCD-eligible families and AISH households is the analytical observation of this chapter.
7. Alberta Health Services Nutrition Guideline on Arsenic in Foods (Chapter II); Health Canada April 2019 announcement (Chapter II); Government of Alberta AISH and ADAP caseload data (this chapter); Food Banks Alberta and Food Banks Canada HungerCount data (Chapter IV); University of Calgary School of Public Policy Kneebone report (Chapter IV). The constructive knowledge doctrine as engaged across Canadian administrative-law and human-rights jurisprudence is documented in the campaign's *Charter and CRPD Analysis* (April 2026).

8. The institutional-cumulative-outcome framing draws on the broader policy-design literature on the cumulative effect of separately optimized administrative frameworks on shared populations. The framing is consistent with the structural-eugenics analysis developed in the campaign's *Consent Was Never the Point* (May 2026), where the intent-claim is distinguished from the effect-claim and the documented argument is the effect-claim.
9. The Alberta Disability System Breakdown, *The Cage Doesn't End* (compilation, May 2026); *Consent Was Never the Point* (May 2026); *The Access Gap: Penalized for Non-Ability* (May 2026). The four-restriction integration is the analytical contribution of this chapter.
10. Cross-reference to *Consent Was Never the Point*, Chapter VII ("Consent Was Never the Point: Five Moments in the Documented Record"), for the campaign's detailed treatment of the historical continuity argument and the distinction between mechanism and architecture.

CLOSING

What This Establishes

What This Establishes

Across the Five Chapters

Five propositions close the piece. They follow from the documented record the five preceding chapters have assembled. They do not require any additional evidentiary or argumentative move beyond what the documented record itself establishes.

The Income Constraint

The maximum monthly living allowance under AISH in 2026 is \$1,940; under ADAP, \$1,740. Against the documented Alberta rent, utility, transportation, and medical out-of-pocket expense structure, the residual budget available to the disabled household for food and household supplies is between zero and approximately three hundred dollars per month. The residual is, in the worked household-budget arithmetic of the campaign's *Income Crossover Analysis* (revised May 2026), insufficient to purchase the food a single adult is documented to require. The residual determines the grocery shelf the household can reach. The grocery shelf the household can reach is the bottom-cost shelf.

The Documented Exposure on the Shelf

The bottom-cost shelf contains, in the Canadian regulatory framework, a documented exposure burden across two categories the comparator regulatory frameworks have moved to restrict, label, or phase out. Healthy Babies Bright Futures' May 2025 testing found inorganic arsenic in 100 percent of 145 rice samples; more than one in four samples exceeded the FDA infant rice cereal limit. Health Canada has not, as of the date of this edition, adopted a maximum level for inorganic arsenic in rice, despite the 2019 commitment to consult toward a Codex-consistent limit and despite the adoption of the 100 parts-per-billion limit by the European Union (2016), the United States (2021), and the Codex Alimentarius Commission (2014, 2016). The seven petroleum-based synthetic dyes the U.S. FDA announced a phase-out of in April 2025 remain on the Canadian permitted list, without the warning-label requirement the European Union has imposed since 2010.

The Food Bank as the Resolution of the Impossibility

The household whose residual budget falls below the documented food cost resolves the impossibility through some combination of three documented mechanisms: cuts to the other expense categories (medication non-adherence, missed medical appointments, reduced utilities), reliance on the food bank, and reliance on familial or community subsidy. The food bank, in the May 22, 2026 University of Calgary School of Public Policy analysis by Dr. Ron Kneebone, is a structural feature of the AISH household budget — not a transitional support the household passes through. Alberta food bank visits have grown by 134.4 percent in the period from March 2019 to March 2025, the largest provincial growth in Canada. The disability cohort is disproportionately represented in the food-bank-use data, and the food supply the food bank distributes draws from the same Canadian food-manufacturing system that supplies the bottom-cost grocery shelf, with the same documented exposure burden.

The Cohort at Which the Burden Concentrates

The 79,290 AISH recipients identified in the September 2025 caseload data are, in 58.5 percent of cases, primary-diagnosed with cognitive or mental-illness presentations. The estimated 16,000 to 20,000 children in AISH households are being raised on the bottom-cost food supply during the developmental window the exposure literature identifies as the period of peak central-nervous-system sensitivity. The body-weight-adjusted exposure across the window is the figure the longitudinal cohort literature has identified as the figure with the largest measurable downstream effects. The cohort least able, on the documented record, to perform the regulatory-literacy task the Canadian ingredient-label disclosure framework assumes is the cohort most structurally concentrated against the products whose labels would, in comparator regulatory frameworks, carry the documented warning. The concentration is the cumulative operational outcome of the four documented frameworks running in parallel on the same population.

The Constructive Knowledge Condition

The Government of Alberta is on documented notice — through its own Alberta Health Services Nutrition Guideline on Arsenic in Foods; through the federal-provincial-territorial Codex consultation record; through its own AISH and ADAP caseload composition data; through the Food Banks Alberta and Food Banks Canada HungerCount surveillance data; through the May 22, 2026 University of Calgary School of Public Policy report — of every element this piece documents. The Government of Canada is on documented notice through its own Health Canada consultation record, through the Codex Alimentarius Commission reference standards, through the United Nations Committee on the Rights of Persons with Disabilities Concluding Observations on Canada of March 18, 2025, and through the published comparator-jurisdiction regulatory records. The continued operation of the integrated framework, in the face of the documented constructive knowledge, is the documented institutional outcome.

The Architecture and the Next Document

This piece does not propose a single regulatory or program reform sufficient to alter the architecture it documents. The architecture's persistence is, on the documented record, the cumulative effect of four separately operated systems (income, food regulation, food manufacturing, food-bank distribution) running in parallel; alteration of the cumulative outcome requires action across all four. The campaign's *Building Up: A Blueprint for the Alberta Accessibility and Inclusion Act* (May 2026) proposes the structural framework — accessibility-and-inclusion legislation surviving any change in government — within which the four-system alteration could be operationalized. The campaign's *Compound Failure Brief*, *Income Crossover Analysis*, *Charter and CRPD Analysis*, *The Access Gap: Penalized for Non-Ability*, and *Consent Was Never the Point* together document the broader architecture this piece sits within.

What the piece is for is the documented record itself. The disabled household in St. Albert, in Edmonton, in Calgary, in Lethbridge, in Camrose, in Red Deer, in Medicine Hat, in Grande Prairie, in Wetaskiwin, in High Level — the household whose monthly food budget the income framework has placed below the documented food cost, whose grocery shelf is the bottom-cost shelf, whose infant or toddler eats the rice cereal the Canadian regulatory framework permits without a maximum arsenic level — that household is the household this piece has been written for. The documented record this piece assembles is the record available, on the campaign's distribution channels, to anyone in that household, in their support network, in their healthcare provider's office, in their MLA's office, in their municipal council's chamber, who would benefit from having the record assembled in a single document.

The food on the shelf is what the income framework has put there. The exposure burden the food carries is what the regulatory framework has permitted. The household that lives on the shelf is the household the AISH and ADAP framework has consigned to it. The architecture is the documented architecture. The record is the documented record.

The mechanism has changed. The architecture has not.

Notes — Closing

1. Cross-references throughout this closing chapter are to the body chapters of this piece (Chapters I through V) and to the campaign's broader documentation library, accessible at albertadisabilitysystembreakdown.netlify.app. The closing chapter does not introduce primary sources not already cited in the body chapters; its purpose is to integrate the documented record assembled in the chapters that precede it.

CLOSING APPARATUS

Bibliography and Sources

Consolidated source apparatus across the five chapters

The bibliography below consolidates the source apparatus across all five chapters of *What's On The Shelf*. Sources are grouped by category. Where the same source appears in multiple chapters, the source is cited once here with the chapters of principal use noted. URLs are provided where applicable. Citations to United Nations, federal, provincial, and academic sources have been verified to their primary records as of the date of this edition.

I. International Standards and Regulatory Instruments

Codex Alimentarius Commission, *General Standard for Contaminants and Toxins in Food and Feed* (CXS 193-1995).

Maximum level for inorganic arsenic in polished rice adopted at the 37th Session, July 2014; maximum level for inorganic arsenic in husked rice adopted at the 39th Session, July 2016. fao.org/fao-who-codexalimentarius/.

Principal use: Ch II.

European Commission, Commission Regulation (EU) 2015/1006 of 25 June 2015 amending Regulation (EC) No 1881/2006 as regards maximum levels of inorganic arsenic in foodstuffs. In force January 1, 2016. Principal use: Ch II.

European Parliament and Council Regulation (EC) No 1333/2008 of 16 December 2008 on food additives, Annex V (warning-label requirement for six synthetic food colours). In force July 20, 2010. Principal use: Ch III.

United Nations Committee on the Rights of Persons with Disabilities, Concluding Observations on the combined second and third periodic reports of Canada, UN Doc CRPD/C/CAN/CO/2-3, adopted 18 March 2025. Principal use: Closing.

U.S. Food and Drug Administration, "Inorganic Arsenic in Rice Cereals for Infants: Action Level," Guidance for Industry, August 2020, finalized 2021. fda.gov. Principal use: Ch II.

U.S. Food and Drug Administration and U.S. Department of Health and Human Services, joint announcement on phase-out of petroleum-based synthetic food dyes, April 22, 2025. Principal use: Ch III.

U.S. Environmental Protection Agency, *Supplemental Guidance for Assessing Susceptibility from Early-Life Exposure to Carcinogens*, EPA/630/R-03/003F, March 2005. Principal use: Ch II, Ch V.

II. Canadian Federal Statutes and Regulations

Canada Disability Benefit Act, SC 2023, c 9. Principal use: Ch IV.

Food and Drug Regulations, CRC c 870 (including Division 16 on permitted food colours). Principal use: Ch III.

Health Canada, *List of Permitted Food Colours* (Lists of Permitted Food Additives).

canada.ca/en/health-canada/services/food-nutrition/food-safety/food-additives/lists-permitted/3-colouring-agents.html, accessed May 2026. Principal use: Ch III.

Health Canada, *Lists of Maximum Levels for Various Chemical Contaminants in Foods*. canada.ca/en/health-canada/services/food-nutrition/food-safety/chemical-contaminants/maximum-levels-chemical-contaminants-foods.html, accessed May 2026. Principal use: Ch II.

Health Canada, *Guidelines for Canadian Drinking Water Quality: Guideline Technical Document — Arsenic*, May 2006 and subsequent updates. Principal use: Ch II.

Health Canada, statement of April 4, 2019, on forthcoming consultation toward Codex-consistent maximum levels for inorganic arsenic in rice (reported in CBC News, April 4, 2019). Principal use: Ch II.

Health Canada, written statement to CBC News, April 23, 2025, on synthetic food dyes (reported in Yoon and Zafar, CBC News, April 23, 2025). Principal use: Ch III.

III. Alberta Statutes, Regulations, and Government Documents

Alberta Health Services, Nutrition Services, *Nutrition Guideline — Healthy Infants and Young Children: Arsenic in Foods*. albertahealthservices.ca/assets/info/nutrition/if-nfs-ng-infants-children-arsenic-in-foods.pdf, accessed May 2026. Principal use: Ch II, Ch V.

Assured Income for the Severely Handicapped General Regulation, AR 96/2026, filed May 12, 2026, in force July 2, 2026. Principal use: Ch I, Ch IV.

Bill 12, Financial Statutes Amendment Act, 2025 (No. 2), SA 2025 c20. Passed December 9, 2025; Royal Assent December 11, 2025. Removed the legislated AISH indexation mechanism. Principal use: Ch IV.

Government of Alberta, AISH Caseload Open Data, September 2025. open.alberta.ca. Principal use: Ch III, Ch IV, Ch V.

Government of Alberta, Family Support for Children with Disabilities (FSCD) program documentation. Principal use: Ch V.

Government of Alberta, AISH and ADAP program documentation, 2025-2026. alberta.ca/alberta-disability-assistance-program. Principal use: Ch I, Ch IV.

IV. Peer-Reviewed and Scholarly Sources

Karagas, M.R., et al., "Association of rice and rice-product consumption with arsenic exposure early in life," *JAMA Pediatrics* 170, no. 6 (2016): 609-616. Principal use: Ch II.

McCann, D., et al., "Food additives and hyperactive behaviour in 3-year-old and 8/9-year-old children in the community: a randomised, double-blinded, placebo-controlled trial," *The Lancet* 370, no. 9598 (3 November 2007): 1560-1567. Principal use: Ch III.

Meharg, A.A., et al., "Geographical variation in total and inorganic arsenic content of polished (white) rice," *Environmental Science and Technology* 43, no. 5 (2009): 1612-1617. Principal use: Ch II.

Williams, P.N., et al., "Variation in arsenic speciation and concentration in paddy rice related to dietary exposure," *Environmental Science and Technology* 39 (2005): 5531-5540. Principal use: Ch II.

European Food Safety Authority, Scientific Opinions on the re-evaluation of food colour additives, 2008-2014, published in the EFSA Journal. Principal use: Ch III.

International Agency for Research on Cancer (IARC), *Monograph on Arsenic*. Group 1 carcinogen classification. Principal use: Ch II.

V. Healthy Babies Bright Futures Reports

Healthy Babies Bright Futures, *Arsenic in 9 Brands of Infant Cereal*, 2017.

hbbf.org/report/arsenic-in-9-brands-of-infant-cereal. Principal use: Ch II.

Healthy Babies Bright Futures, *What's in My Baby's Food?*, October 2019. hbbf.org. Tested 168 baby foods; 95 percent positive for at least one heavy metal. Principal use: Ch II, Ch V.

Healthy Babies Bright Futures, *What's in Your Family's Rice?*, May 15, 2025. hbbf.org/report/arsenic-in-rice. Tested 145 rice samples; 100 percent contained arsenic; more than 1 in 4 exceeded FDA infant rice cereal limit. Principal use: Ch II.

Healthy Babies Bright Futures, *Arsenic in Infant Formula and Toddler Foods*, 2012. Principal use: Ch II.

VI. Food Bank and Income Support Documentation

Food Banks Canada, HungerCount 2024 national report, October 2024. foodbankscanada.ca. Principal use: Ch IV.

Food Banks Canada, Alberta 2024 Report Card, October 28, 2024.

foodbankscanada.ca/poverty-report-card/report-card/?p=alberta&y=2024. Principal use: Ch IV.

Food Banks Alberta, 2021 HungerCount Findings for Alberta (Arianna Scott, Interim CEO), November 2021. Principal use: Ch IV.

Kneebone, R., et al., University of Calgary School of Public Policy, report on AISH households and food bank use, May 22, 2026. Reported in Medicine Hat News, May 22, 2026. Principal use: Ch IV, Ch V, Closing.

Maytree Foundation, Social Assistance Summaries, Alberta, March 31, 2026 update.

maytree.com/changing-systems/data-measuring/social-assistance-summaries/alberta/. Principal use: Ch IV.

PROOF Food Insecurity Policy Research, University of Toronto. proof.utoronto.ca. Principal use: Ch IV.

VII. Statistical and Survey Sources

Canada Mortgage and Housing Corporation, Rental Market Report 2024. cmhc-schl.gc.ca. Principal use: Ch I.

Dalhousie University, University of British Columbia, University of Saskatchewan, and University of Guelph, *Canada's Food Price Report 2026*, December 2025. Principal use: Ch I.

Statistics Canada, monthly food cost estimates for a single adult, 2026. Principal use: Ch I.

Statistics Canada, Consumer Price Index data on food prices, 2020-2026. Principal use: Ch I, Ch III.

Urbanation / Rentals.ca, National Rent Report, June 2024. Principal use: Ch I.

VIII. The Alberta Disability System Breakdown Campaign Documentation

The Alberta Disability System Breakdown, *Access Barrier Report*, April 2026. Principal use: Ch I, Ch II.

The Alberta Disability System Breakdown, *The Access Gap: Penalized for Non-Ability*, May 2026. Principal use: Ch III, Ch V.

The Alberta Disability System Breakdown, *AISH Evidentiary Report Series*, April 2026. Principal use: throughout.

The Alberta Disability System Breakdown, *Building Up: A Blueprint for the Alberta Accessibility and Inclusion Act*, May 2026. Principal use: Closing.

The Alberta Disability System Breakdown, *The Cage Doesn't End* compilation, May 2026. Principal use: Ch I, Ch V.

The Alberta Disability System Breakdown, *Charter and CRPD Analysis*, April 2026. Principal use: Ch V.

The Alberta Disability System Breakdown, *Compound Failure Brief*, April 2026. Principal use: Ch IV, Closing.

The Alberta Disability System Breakdown, *Consent Was Never the Point*, May 2026. Principal use: Ch V.

The Alberta Disability System Breakdown, *Income Crossover Analysis*, revised May 2026. Principal use: Ch I, Ch IV, Closing.

albertadisabilitysystembreakdown.netlify.app. Campaign website. Distribution platform for all of the documents in this section.

IX. News Media and Investigative Reporting

CBC News, "Health Canada says it will set cap on arsenic in food," April 4, 2019.
cbc.ca/news/health/arsenic-rice-baby-food-1.5086078. Principal use: Ch II.

CBC News, "Number of Albertans relying on income support surges to 57,000," January 6, 2025.
cbc.ca/news/canada/edmonton/alta-income-support-1.7423736. Principal use: Ch IV.

Medicine Hat News, "Report says Albertans on disability supports disproportionately use food banks," May 22, 2026.
medicinehatnews.com. Principal use: Ch IV.

Yoon, J. and Zafar, A., "RFK Jr. calls artificial food dyes 'poisonous.' Here's how they're regulated in Canada," CBC News, April 23, 2025. cbc.ca/news/health/food-dye-1.7516794. Principal use: Ch III.

The bibliography above is current to May 2026. Subsequent editions of this piece will incorporate corrections, additions, and updates as the documented record develops. The campaign maintains the source apparatus in machine-readable form at albertadisabilitysystembreakdown.netlify.app. Corrections may be directed to albertadisabilitybreakdown@outlook.com.

ON THE RECORD · MAY 2026

This piece is published by The Alberta Disability System Breakdown, an independent public advocacy campaign documenting the mandatory transition of approximately 79,290 Albertans from the Assured Income for the Severely Handicapped (AISH) program to the Alberta Disability Assistance Program (ADAP), effective July 1, 2026, alongside the broader institutional and legislative environment in which the transition has been undertaken.

The campaign operates as a documentation-first public record. Citations are provided throughout. The principal academic, regulatory, and primary-source references are listed in the bibliography that precedes this page and have been verified to their primary records. The campaign maintains the full source apparatus in machine-readable form at albertadisabilitysystembreakdown.netlify.app.

Distribution

This piece may be freely quoted, reproduced, and redistributed under the terms of attribution to The Alberta Disability System Breakdown. No individual byline is asserted. The campaign holds the work as collective record. Corrections, additions, and verification notes may be directed to albertadisabilitybreakdown@outlook.com.

The food on the shelf is what the income framework has put there. The exposure burden the food carries is what the regulatory framework has permitted. The architecture is the documented architecture.

The Alberta Disability System Breakdown — Advocate

St. Albert, Alberta | May 2026

albertadisabilitybreakdown@outlook.com

Free to share · Free to print · Free to post